

TrayGuard

A Local Tray Training and Assessment System

Elevator Pitch — Final Design Review

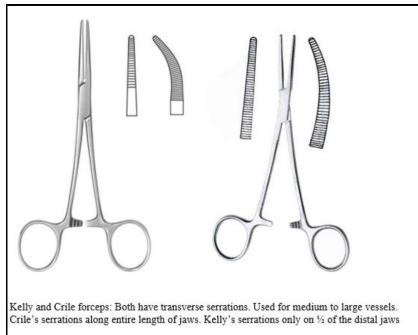
The Problem

- **9 defects per 100** surgical cases
- **55%** occur during tray assembly
- **10 min** average delay per affected case
- **\$6.75–9.42M** estimated annual cost per hospital
- **44%** of packaging errors: wrong specification

“The total count was correct. The specific instrument was wrong.”

Root cause

Visual discrimination under pressure — telling similar instruments apart at a glance.



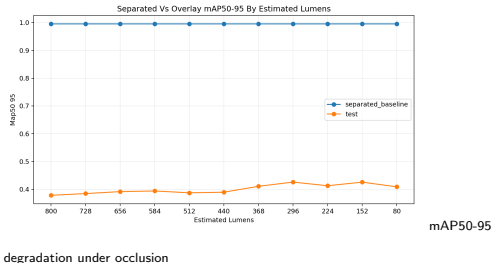
Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels. Crile's serrations along entire length of jaws. Kelly's serrations only on ½ of the distal jaws

same family, different serration pattern

Kelly vs Crile —

Why Automation Failed

- Seven experiments: brightness, background, layout occlusion
- Every real-world condition **degraded detection**
- Occlusion: **mAP50-95 dropped to 0.389**
- The model learned **position, not shape**
- Site-specific data + validation needed per hospital
- A camera cannot fix a broken feedback loop



Decision

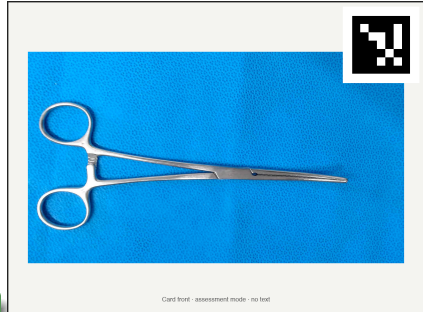
Deployment-grade recognition requires more data and institutional adoption than one project can responsibly deliver.

Introducing TrayGuard

- Training platform: **laptop + printer**
- Printable instrument cards with AprilTag markers
- 7-step evidence-backed learning loop
- Scoring with 6 error categories
- Confidence capture per item
- Pre/post metrics export

What it is not

TrayGuard does not replace certification. It closes the gap between certification and local tray readiness.



Clamp Kelly 8"

Family: Clamping/occluding · Catalog: CLAMP KELLY 8"

Key Identification features

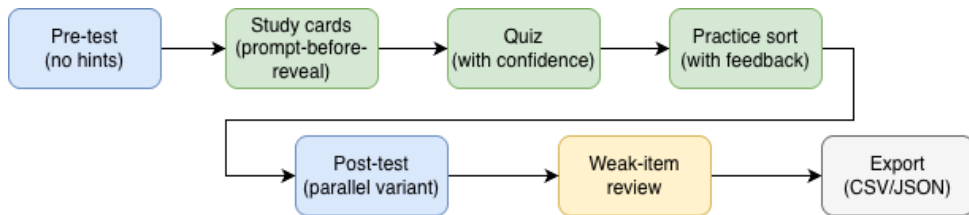
- Curved jaws with ring handles
- Transverse serrations on distal half of jaws
- 8-inch overall length

Common lookalikes

- Crile clamp — serrations run full jaw (Kelly: distal half only)
- Kocher clamp — single tooth at tip (Kelly: no tooth)
- Kelly 6" / Kelly 12" — same jaw shape, different length

Also known as: JARIT_106-191

The Learning Loop



Pre / post use parallel variants. Study cards are retrieval practice. Practice gives error-specific feedback.

Pre / Post

Parallel variants
No hints
Baseline + gain

Study & Quiz

Retrieval practice
Immediate feedback
Confidence capture

Practice sort

Physical card placement
Error-category feedback
Weak-item review

What It Captures

Category	Maps to literature
Missing	Alfred et al.
Extra	Alfred et al.
Wrong item	Alfred, Nichol
Misidentified	Zhu — 44% wrong spec
Wrong count	Zhu
Correct	—

Confidence: How sure are you?

Rate your confidence for each tray selection:

1

2

3

4

5

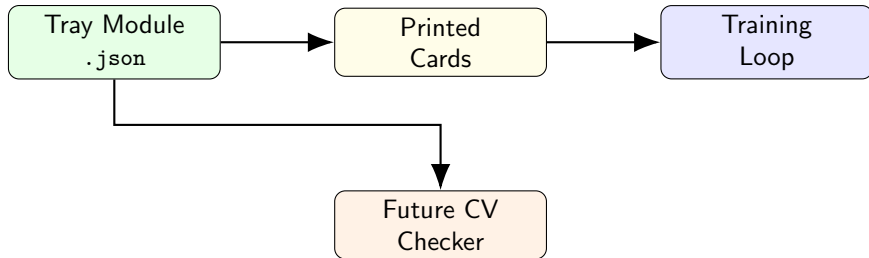
Guessing Unsure Neutral Fairly sure Certain

Not Sure

Confidence ratings separate:

- High-confidence errors — “confidently wrong”
- Low-confidence correct — fragile knowledge

The Durable Artifact



Local knowledge is prerequisite infrastructure

The module captures names, aliases, lookalikes, and counts — needed whether you are training now or automating later.

Current Status & Next Steps

Delivered

- Problem evidence from published literature
- Working web app with 7-step learning loop
- 8 file-backed tray modules
- Scoring engine with 6 categories
- Card generation pipeline
- Study protocol with falsification criteria

Next step: pilot

- Baseline pre/post study (8–12 participants)
- 7-day retention check
- Expert content review

Seeking partner

Hospital or SPD training program to pilot the first module with real tray content.